

Implementation of an Authoring Tool for Wheelchair Simulation with Visual and Vestibular Feedback

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ABSTRACT

In this study, we develop a wheelchair simulator that provides visual and vestibular feedback at a low cost by leveraging the combination of vection-inducing movies displayed on a head-mounted display and vestibular feedback provided by an electric-powered wheelchair. However, this simulator requires users to manually create a synchronized pair of a VR movie and a motion scenario (chronological control commands for the wheelchair), which necessitates considerable effort and experience. In this paper, we introduce a novel authoring tool that generates a pair of a VR movie and a motion scenario based on a few parameters entered by the user.

Index Terms: Human-centered computing—Accessibility—Accessibility system and tools

1 INTRODUCTION

VR-based wheelchair simulators (VRWSs) have been used for rehabilitation and training because they can provide users with a controllable, repeatable, and safe training environment [1, 4]. However, typical VRWSs exhibit a trade-off between a cost and reality. Simulators that provide only visual feedback can be constructed at lower costs but cannot provide users with a sense of motion. Meanwhile, simulators that provide both visual and vestibular feedback can yield richer experiences but require expensive devices. Hence, we have developed a VRWS that provides visual and vestibular feedback at a low cost. It involves vection-inducing movies displayed on a head-mounted display (HMD) for enhancing vestibular feedback provided by a programmable electric-powered wheelchair (EPW) [3]. In this paper, we introduce a novel authoring tool for this simulator. When the user inputs the attributes of a barrier to be simulated, this tool outputs a VR movie from a first-person view and a motion scenario (EPW control commands in chronological order) simultaneously.

2 RELATED WORK

Our study pertains to studies on VRWSs, as well as authoring tools for VR environments.

2.1 VR-based wheelchair simulators

VRWSs can be broadly classified into two groups [1, 4]: simulators that provide only visual feedback and simulators that provide both visual and vestibular feedback.

VRWSs that provide only visual feedback [5, 6] afford a simple system structure. The basic components include a chair on which the user is to sit, a display (including a normal display [5] and an HMD [6]) that presents the virtual environment, and a joystick that allows the user to maneuver within the environment. The simplicity of this configuration allows cost savings; however, this approach cannot provide users with vestibular feedback.

VRWSs that provide both visual and vestibular feedback [2, 8] enable users to have a richer experience. However, for providing

vestibular feedback, a motion platform with 6DoF, known as the Stewart platform, is typically used; it is expensive and requires a large space for permanent installation.

2.2 Authoring tools for VR environments

Commercial applications (e.g., Unity ¹, Unreal Engine ², Substance 3D Stager ³, etc.) for constructing VR environments abound. These applications allow users to manually create VR courses for wheelchair training. Some studies focused on the automatic creation of VR environments by scanning physical objects [7] and real environments [9]. This approach allows users to intuitively create VR courses.

3 RESEARCH GOAL

Simulators that provide only visual feedback can be constructed at lower costs but cannot provide users with a sense of motion. On the other hand, simulators that provide both visual and vestibular feedback can yield richer experiences but require expensive devices. To overcome the trade-off between cost and reality, we have developed a VRWS that provides visual and vestibular feedback using a combination of vection-inducing movies displayed on an HMD and motions attainable by a programmable EPW ⁴ [3] (Figure 1). This simulator requires a *VR movie* and a *motion scenario*. The VR movie reproduces the scene of moving the virtual course in a wheelchair from a first-person perspective. The motion scenario is a series of EPW control commands that define the speed and the direction of the wheelchair at each moment. The motion scenario is stored in a single-board computer connected to the EPW, which sends control commands to the EPW. The VR movie and the motion scenario must be synchronized, i.e., the same event must appear simultaneously in both the VR movie and the motion scenario.

Currently, VR movies and motion scenarios are created and synchronized manually, which requires considerable effort and experience. To the best of our knowledge, no VR authoring tool specifically for creating wheelchair training courses is available. Although scanning approaches [7, 9] may reduce the effort of CG modeling, they require users to go to actual places and scan the shape of barriers, such as steps and slopes. Furthermore, these VR authoring tools do not support the creation of motion scenarios.

Hence, we conducted this study to develop of a simulation authoring tool that allows users to create synchronized VR movies and motion scenarios for our VRWS with less effort and experience.

4 SYSTEM DESIGN

To achieve the research goal, we designed the concept of the simulation authoring tool, as shown in Figure 2. This tool accepts barrier shape information as input, thereby reducing the effort for the manual modeling of CG objects. Based on the entered barrier shape information, this tool generates a pair of a VR movie and a motion scenario. The VR movie and the motion scenario are synchronized: the shape of the barrier is determined based on the barrier shape

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¹<https://unity.com>

²<https://www.unrealengine.com/>

³<https://www.adobe.com/products/substance3d-stager.html>

⁴WHILL, <https://whill.inc/us>

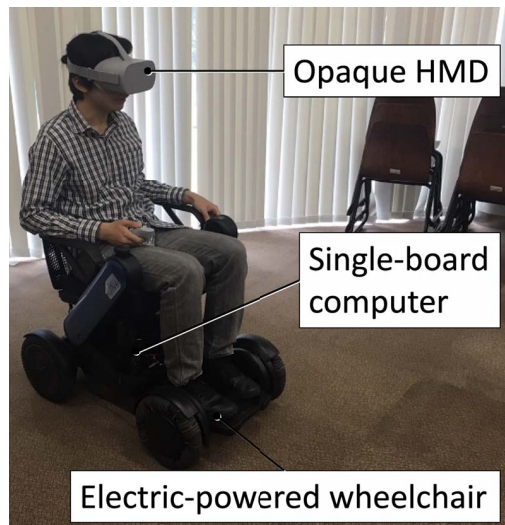


Figure 1: Appearance of proposed simulator.

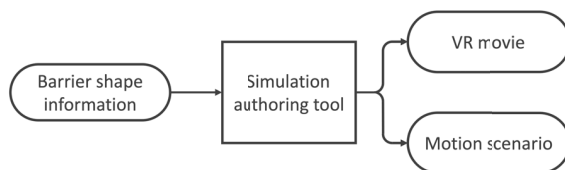


Figure 2: Concept of simulation authoring tool.

information, and EPW control commands are determined based on the barrier shape information and the effect of the gravitational acceleration. We assume that this tool will be used on a PC by people who will experience the simulation.

5 IMPLEMENTATION

For the first step, we focus on up- and down-slope simulations [3]. Our VRWS displays an up-slope on the HMD and decelerates the EPW for the up-slope simulation. Conversely, the VRWS displays a down-slope on the HMD and accelerates the EPW for the down-slope simulation.

To generate a pair of a VR movie and a motion scenario for slope simulation, the authoring tool accepts slope attributes as input; *direction (up or down)*, *slope length*, *slope angle*, and *lengths of the flat roads before and after the slope*. Subsequently, the tool generates a VR movie that reproduces the scene of passing through the slope in a wheelchair from a first-person view (Figure 3 (top left and bottom left)). Moreover, this tool renders the CG of the slope from a bird's-eye view (Figure 3 (top right and bottom right)) such that the user can view the entire picture of the slope. Simultaneously, this tool generates a motion scenario that considers the shape of the slope and the effect of gravitational acceleration. Specifically, the EPW speed decreases at the start of the up-slope, whereas it increases at the start of the down-slope. In this study, the slope angle was assumed to be constant.

6 CONCLUSION

We proposed a novel simulation authoring tool to support the creation of simulation contents for wheelchair simulators that leverage the combination of vection-inducing movies and EPW motions. This tool generates a synchronized pair of a VR movie and a motion scenario based on the entered barrier shape information. In future work,

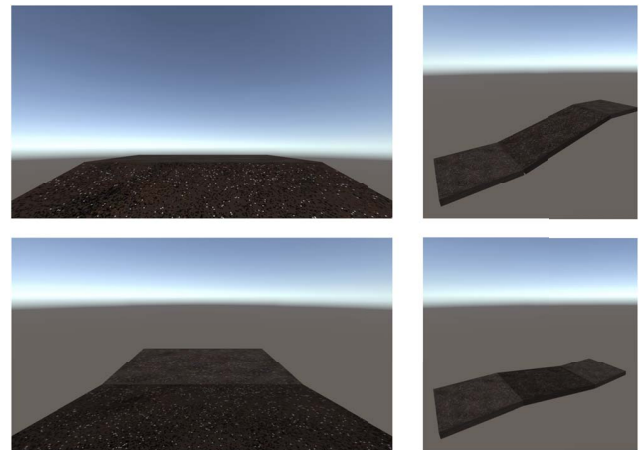


Figure 3: Left: VR movies of generated slope from first-person view (top left: up-slope; bottom left: down-slope). Right: Entire generated slope from bird's-eye view (top right: up-slope; bottom right: down-slope).

we plan to evaluate the usability of this tool and the quality of the generated simulation contents. Furthermore, we plan to support various barrier types other than slopes.

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